

Computer Aided Geometric Design

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B-spline Curves

Laurent Busé

Centre Inria d'Université Côte d'Azur,
Email : laurent.buse@inria.fr



B-spline Curves - Motivation

- ▶ **Most shapes are too complicated to define using a single Bézier curve.**
- ▶ A **spline curve is a sequence of Bézier curves** that are connected together
- ▶ $\mathcal{C}^1$ continuity is easy to attain with Bézier curves, but $\mathcal{C}^2$ and higher continuity is cumbersome.

B-spline curves

This is a method for representing a sequence of degree n Bézier curves that join automatically with $\mathcal{C}^{n-1}$ continuity, regardless of where the control points are placed.

- ▶ An open string of m Bézier curves of degree n involves $nm + 1$ distinct control points,
- ▶ We will see the same string of Bézier curves can be expressed using only $m + n$ B-spline control points (assuming $\mathcal{C}^{n-1}$ continuity).

Polar Forms - Motivation

- ▶ A degree n polynomial can be written in several bases:

$$f(t) = \sum_{i=0}^n a_i t^i \quad (\text{monomial}), \quad f(t) = \sum_{i=0}^n c_i B_i^n(t) \quad (\text{Bernstein}).$$

- ▶ In each basis, the coefficients are unique and determine a unique polynomial function.
- ▶ If the coefficients are points in $\mathbb{R}^p$, we obtain parametric polynomial curves in $\mathbb{R}^p$; for instance, Bézier curves with $c_i \in \mathbb{R}^2$ or $\mathbb{R}^3$ and $t \in [0, 1]$.

Key idea

Give intrinsic labels to the coefficients of a univariate polynomial (equivalently to control points of a polynomial curve), independently of the chosen basis.

Polar Forms - Definition

Definition (Polar form)

Let $f : \mathbb{R} \rightarrow \mathbb{R}^p$ be a polynomial function of degree at most n . A map

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^p, \quad (t_1, \dots, t_n) \mapsto F(t_1, \dots, t_n)$$

is called the *polar form* of f if:

- ▶ $F(t, \dots, t) = f(t)$ for all $t \in \mathbb{R}$,
- ▶ F is symmetric in its arguments,
- ▶ F is multi-affine.

Multi-affine

For each argument, affine combinations are preserved. For example,

$$F(\dots, (1 - \lambda)u + \lambda v, \dots) = (1 - \lambda)F(\dots, u, \dots) + \lambda F(\dots, v, \dots).$$

Polar Forms - Existence and Uniqueness

Theorem (Polar form theorem)

Every degree n polynomial map $f : \mathbb{R} \rightarrow \mathbb{R}^p$ has a unique polar form.

Proof - existence

If

$$f(t) = \sum_{i=0}^n a_i t^i,$$

then a polar form is

$$F(t_1, \dots, t_n) = \sum_{i=0}^n a_i \frac{\sigma_i(t_1, \dots, t_n)}{\binom{n}{i}},$$

where σ_i is the elementary symmetric polynomial of degree i :

$$\sigma_i(t_1, \dots, t_n) = \sum_{1 \leq j_1 < \dots < j_i \leq n} t_{j_1} \cdots t_{j_i}, \quad \sigma_0 = 1.$$

Polar Forms - Existence and Uniqueness

Proof - degrees of freedom

Claim: F is completely determined by the $n + 1$ values

$$F_i := F(\underbrace{0, \dots, 0}_{n-i}, \underbrace{1, \dots, 1}_i), \quad i = 0, \dots, n.$$

Indeed, since F is symmetric, multi-affine and $t_1 = (1 - t_1).0 + t_1.1$,

$$\begin{aligned} F(t_1, 0, \dots, 0, 1, \dots, 1) = \\ (1 - t_1)F(0, 0, \dots, 0, 1, \dots, 1) + t_1F(0, \dots, 0, 1, \dots, 1, 1). \end{aligned}$$

Therefore, recursively:

$$\begin{aligned} F(t_1, \dots, t_r, 0, \dots, 0, 1, \dots, 1) = \\ (1 - t_r)F(t_1, \dots, t_{r-1}, 0, 0, \dots, 0, 1, \dots, 1) + t_rF(t_1, \dots, t_{r-1}, 0, \dots, 0, 1, \dots, 1, 1). \end{aligned}$$

Polar Forms - Existence and Uniqueness

Proof - uniqueness

It remains to show that $f(t) = F(t, \dots, t)$ determines the values

$$F_i = F(\underbrace{0, \dots, 0}_{n-i}, \underbrace{1, \dots, 1}_i).$$

We proceed similarly to the previous slide:

$$F(t, \dots, t) = (1 - t)F(0, t, \dots, t) + tF(1, t, \dots, t),$$

then,

$$F(0, t, \dots, t) = (1 - t)F(0, 0, t, \dots, t) + tF(0, 1, t, \dots, t), \quad \text{etc.}$$

In the end, we get

$$f(t) = F(t, \dots, t) = \sum_{i=0}^n F_i \binom{n}{i} (1 - t)^{n-i} t^i = \sum_{i=0}^n F_i B_i^n(t).$$

Polar Values and Bernstein Coefficients

Theorem (Bernstein coefficients from polar values)

If $f(t) = \sum_{i=0}^n c_i B_i^n(t)$, then

$$c_i = F(\underbrace{0, \dots, 0}_{n-i}, \underbrace{1, \dots, 1}_i), \quad i = 0, \dots, n.$$

Example

For $f(t) = 1 + t + t^2$ and $n = 2$,

$$F(t_1, t_2) = 1 + \frac{t_1 + t_2}{2} + t_1 t_2.$$

Hence

$$(c_0, c_1, c_2) = \left(1, \frac{3}{2}, 3\right), \quad f(t) = B_0^2(t) + \frac{3}{2}B_1^2(t) + 3B_2^2(t).$$

Polar Values and Bézier Curves

- ▶ A Bézier curve $P(t) = \sum_{i=0}^n P_i B_i^n(t)$ corresponds to a unique polar form $P(t_1, \dots, t_n)$
- ▶ Control points can be identified with **polar values**.

Control points of a Bézier curve

The control points of a degree n Bézier curve $\mathbf{P}_{[a,b]}(t)$ are relabeled as

$$\mathbf{P}_i := \mathbf{P}(u_1, u_2, \dots, u_n), \quad u_1 = \dots = u_{n-i} = a, \quad u_{n-i+1} = \dots = u_n = b.$$



For a degree-3 Bézier curve, the polar values are

$$\mathbf{P}_0 = \mathbf{P}(a, a, a),$$

$$\mathbf{P}_1 = \mathbf{P}(a, a, b),$$

$$\mathbf{P}_2 = \mathbf{P}(a, b, b),$$

$$\mathbf{P}_3 = \mathbf{P}(b, b, b).$$

Subdivision of Bézier Curves Revisited

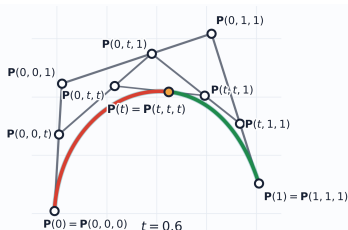
To illustrate how polar values work, we show how to derive the de Casteljau algorithm using only the first three rules.

- ▶ Given a cubic Bézier curve $\mathbf{P}_{[0,1]}(t)$, we wish to split it into $\mathbf{P}_{[0,t]}$ and $\mathbf{P}_{[t,1]}$.
- ▶ The original control points are given by the polar values

$$\mathbf{P}(0, 0, 0), \mathbf{P}(0, 0, 1), \mathbf{P}(0, 1, 1), \mathbf{P}(1, 1, 1).$$

and we need to compute the polar values

$$\begin{array}{cccc} \mathbf{P}(0, 0, 0), & \mathbf{P}(0, 0, t), & \mathbf{P}(0, t, t), & \mathbf{P}(t, t, t) \\ \mathbf{P}(t, t, t), & \mathbf{P}(t, t, 1), & \mathbf{P}(t, 1, 1), & \mathbf{P}(1, 1, 1). \end{array}$$



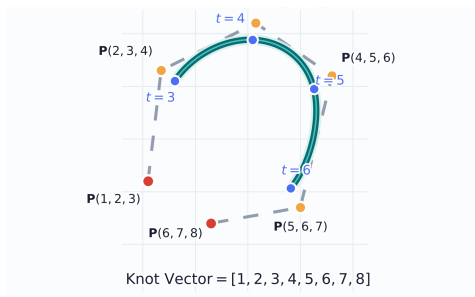
Corollary

$$\forall t : \mathbf{P}(t, t, t) = \mathbf{P}(t)$$

B-spline Curve Definition

A B-spline curve is defined by the following data:

- ▶ a **degree**: $n \geq 1$,
- ▶ a **knot vector** $[t_1, t_2, t_3, \dots]$ such that $t_1 \leq t_2 \leq t_3 \leq \dots$,
- ▶ control points which are **polar values at n adjacent knots** from the knot vector, with the i -th polar value being $\mathbf{P}(t_i, t_{i+1}, \dots, t_{i+n-1})$.



In the next slides, we will explain how the curve is defined from this data.

Knot Vectors

- ▶ A knot vector is a non-decreasing sequence of parameter values.
- ▶ It specifies the parameter intervals for the individual Bézier curves that make up a B-spline.

Example

If a cubic B-spline is made of four Bézier curves with parameter intervals $[1, 2]$, $[2, 4]$, $[4, 5]$ and $[5, 8]$, then the knot vector would be

$$[t_1, t_2, 1, 2, 4, 5, 8, t_8, t_9].$$

There are two (one less of the degree) extra knots appended and prepended to the knot vector. They control the *end conditions* of the B-spline (see later).

Terminology:

- ▶ If any knot is repeated, it is referred to as a *multiple knot*.
- ▶ A B-spline whose knots are evenly spaced is known as a **uniform** B-spline.

Bézier curves as B-splines

- ▶ The *standard* knot vector of a degree n Bézier curve is $[0, \dots, 0, 1, \dots, 1]$, which is of size $2n$.
- ▶ A nondegenerate knot vector $[u_1, \dots, u_n, u_{n+1}, \dots, u_{2n}]$ also defines a Bézier curve $P(t)$, with $t \in [u_n, u_{n+1}]$.

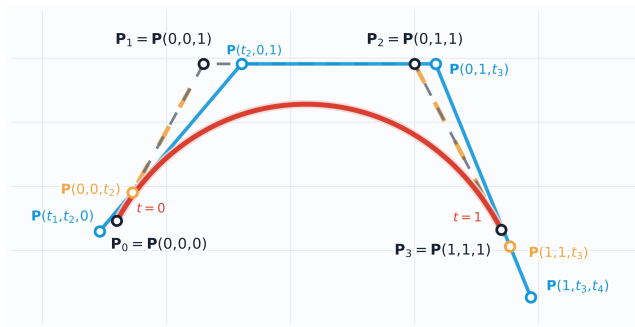


Figure: Bézier and B-spline control polygons of the same cubic Bézier curve, for the knot vectors $[0, 0, 0, 1, 1, 1]$ and $[t_1, t_2, 0, 1, t_3, t_4]$.

The idea behind data of B-spline curves

Main concept

Suppose given a degree n B-spline curve with the knot vector

$$[u_1, \dots, u_{2n}, u_{2n+1}, u_{2n+2}, \dots, u_m].$$

Each sequence of $2n$ consecutive knots provides data for a Bézier curve.

- $[u_1, \dots, u_{2n}]$ defines a Bézier curve on the interval $[u_n, u_{n+1}]$,
- $[u_2, \dots, u_{2n+1}]$ defines a Bézier curve on the interval $[u_{n+1}, u_{n+2}]$,
- etc.

Numerology:

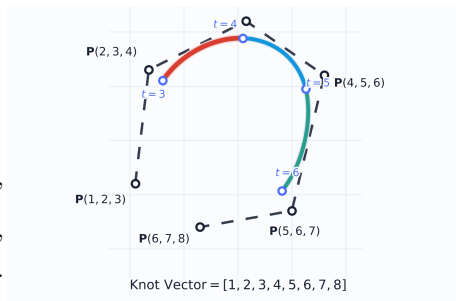
- ▶ The B-spline curve contains $m - 2n + 1$ Bézier curve segments.
- ▶ The addition of a Bézier segment is done by adding one control point.

Extracting Bézier Curves from B-splines

Böhm algorithm

How does one find the control points of the Bézier curves that make up a B-spline?

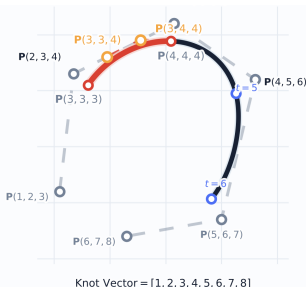
- ▶ Consider the B-spline below consisting of three Bézier curves in the domains $[3, 4]$, $[4, 5]$ and $[5, 6]$.
- ▶ The control points of these Bézier curves have polar values
 $P(3, 3, 3)$, $P(3, 3, 4)$, $P(3, 4, 4)$, $P(4, 4, 4)$,
 $P(4, 4, 4)$, $P(4, 4, 5)$, $P(4, 5, 5)$, $P(5, 5, 5)$,
 $P(5, 5, 5)$, $P(5, 5, 6)$, $P(5, 6, 6)$, $P(6, 6, 6)$.
- ▶ Goal : compute these polar values.



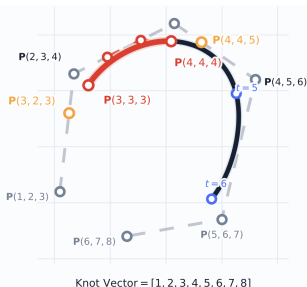
Extracting Bézier Curves from B-splines

For the Bézier curve over $[3, 4]$,

- ▶ First step: compute $\mathbf{P}(3, 3, 4)$ and $\mathbf{P}(3, 4, 4)$.
- ▶ Second step: compute the auxiliary points $\mathbf{P}(3, 2, 3)$ and $\mathbf{P}(4, 4, 5)$.
- ▶ Finally, compute $\mathbf{P}(3, 3, 3)$ and $\mathbf{P}(4, 4, 4)$.



(a) First step



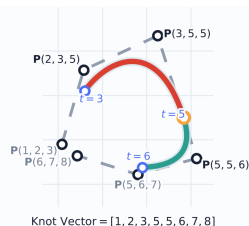
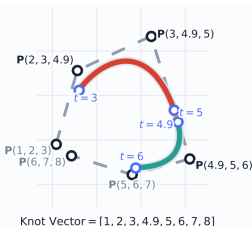
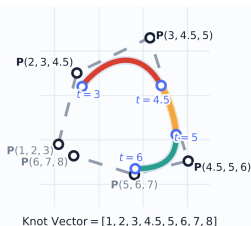
(b) Second step

B-splines possess the property of *local control*

The four Bézier control points are derived from four B-spline control points; $\mathbf{P}(5, 6, 7)$ and $\mathbf{P}(6, 7, 8)$ do not affect this Bézier segment.

Multiple knots

- ▶ If a knot vector contains two identical non-end-condition knots $t_i = t_{i+1}$, the B-spline can be thought of as containing a zero-length Bézier curve over $[t_i, t_{i+1}]$.
- ▶ Below is an illustration of two points which are moved together.



A multiple knot diminishes the continuity

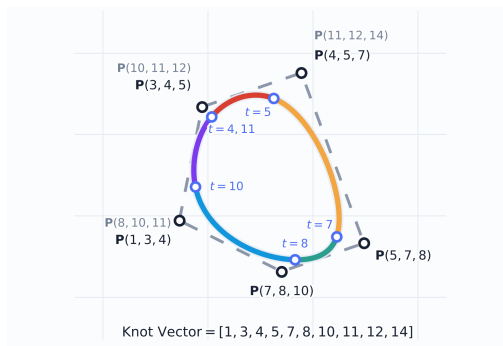
The continuity of two adjacent Bézier curves across a knot of multiplicity k is generally $n - k$.

Periodic B-splines

A periodic B-spline closes with matching derivatives at the join.

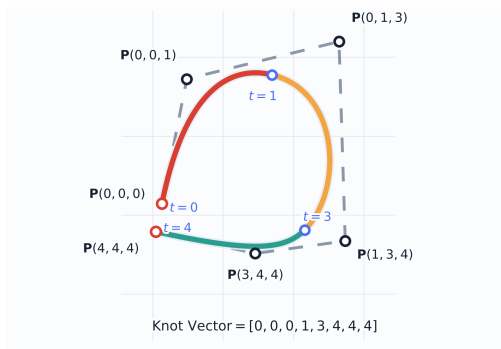
For a degree n B-spline curve with simple knots:

- ▶ require equal endpoint values, $\mathbf{P}(a) = \mathbf{P}(b)$ (C^0 closure);
- ▶ repeat the first n control points at the end and extend the knot sequence by translation through one period (C^{n-1} join).



Bézier end conditions

- ▶ We already noted that a knot vector always has $n - 1$ extra knots at the beginning and end which do not specify Bézier parameter limits (except in the periodic case), but which influence the shape of the curve at its endpoints.
- ▶ In the case of an open B-spline, one usually chooses an n -fold knot at each end to get a Bézier behavior: the curve interpolates the end control points and is tangent to the control polygon at its endpoints (Böhm algorithm).

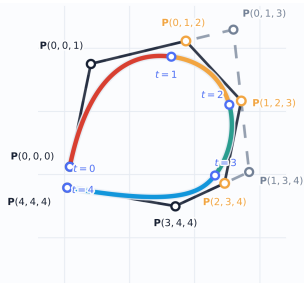


Knot insertion

- ▶ It is a standard design tool for B-splines consisting of adding a knot in the knot vector.
- ▶ It results in an additional control point and a modification of a few existing ones. The curve itself is unchanged.
- ▶ Several applications: evaluation, add local details, *splitting in Bézier segments*, etc.

In the example below a knot is inserted at $t = 2$.

	Initial	After insertion
Knot vector:	$[0, 0, 0, 1, 3, 4, 4, 4]$	$[0, 0, 0, 1, 2, 3, 4, 4, 4]$
Control points:	$P(0, 0, 0)$	$P(0, 0, 0)$
	$P(0, 0, 1)$	$P(0, 0, 1)$
	$P(0, 1, 3)$	$P(0, 1, 2)$
	$P(1, 3, 4)$	$P(1, 2, 3)$
	$P(3, 4, 4)$	$P(2, 3, 4)$
	$P(4, 4, 4)$	$P(3, 4, 4)$
		$P(4, 4, 4)$



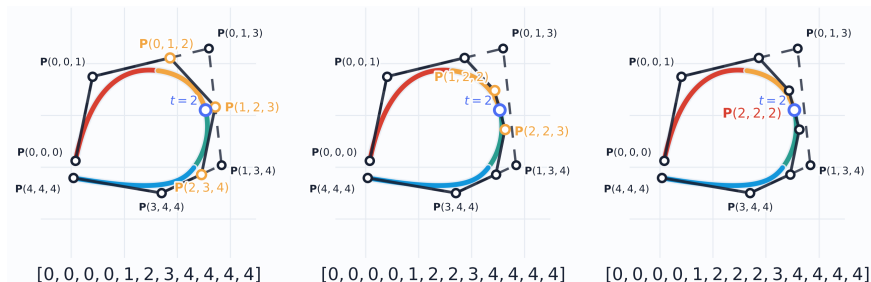
The de Boor algorithm

The de Boor algorithm provides a method for evaluating a B-spline curve.

- ▶ Given a parameter value t , find the corresponding point on the B-spline.
- ▶ This point $\mathbf{P}(t)$ has the polar value $\mathbf{P}(t, t, \dots, t)$.
- ▶ It can be computed by inserting n times the knot t .
- ▶ Using polar forms, the algorithm is easy to figure out.

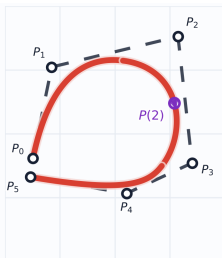
Initial knot vector $[0, 0, 0, 0, 1, 3, 4, 4, 4, 4]$

Goal: compute $\mathbf{P}(2, 2, 2)$.

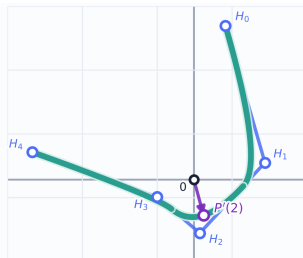


B-spline hodographs

- ▶ The first derivative (or hodograph) of a B-spline is obtained as for Bézier curves.
- ▶ Its knot vector is the original one with the first and last knots discarded.
- ▶ Its control points are $H_i := \frac{n}{t_{i+n} - t_i} (\mathbf{P}_{i+1} - \mathbf{P}_i)$, where n is the degree.



Cubic B-spline, knots [0, 0, 0, 1, 3, 4, 4, 4]



Hodograph, knots [0, 0, 1, 3, 4, 4]

B-spline Curves from Basis Functions

Property

A degree n B-spline curve with $m + 1$ control points can be expressed as

$$\mathbf{P}(t) = \sum_{i=0}^m \mathbf{P}_i N_i^n(t).$$

The functions $N_i^n(t)$ are called the *B-spline basis functions* (to be defined).

Using a geometric point of view, we have seen that:

- ▶ The curve has a knot vector $[t_1, \dots, t_{m+n}]$.
- ▶ The control point $\mathbf{P}_i$ has polar value $\mathbf{P}(t_{i+1}, \dots, t_{i+n})$.
- ▶ This curve is composed of $m - n + 1$ Bézier curves corresponding to the parameter intervals $[t_{n+j-1}, t_{n+j}]$, $j = 1, \dots, m - n + 1$.

B-spline Basis Functions

N.B.: For basis functions, the (reduced) knot vector used for polar values must be completed (by duplicating the first and last knots); the knot vector becomes:

$$[\mathbf{t}_0, t_1, \dots, t_{m+n}, \mathbf{t}_{m+n+1}].$$

Definition

The functions $N_i^0(t)$ are the step functions; they are defined as

$$N_i^0(t) = \begin{cases} 1 & \text{if } t \in [t_i, t_{i+1}[\\ 0 & \text{otherwise} \end{cases}$$

Then, B-spline basis functions of degree k are defined using the recurrence relation

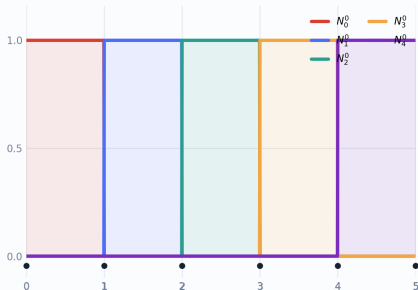
$$N_i^k(t) = w_i^k(t)N_i^{k-1}(t) + (1 - w_{i+1}^k(t))N_{i+1}^{k-1}(t)$$

where

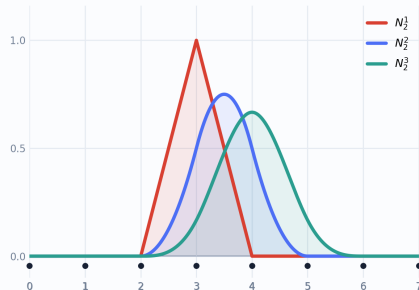
$$w_i^k(t) = \begin{cases} \frac{t-t_i}{t_{i+k}-t_i} & \text{if } t_i < t_{i+k} \\ 0 & \text{otherwise} . \end{cases}$$

B-spline Basis Functions: Examples

Degree 0: step functions

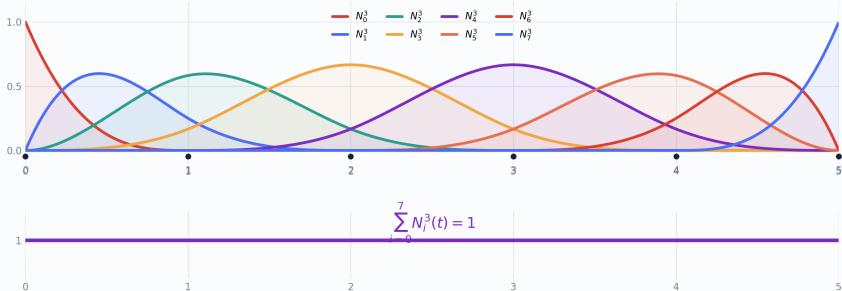


Uniform knots: increasing degree



B-spline Basis: Partition of Unity

Cubic B-spline basis functions



Partition of unity

The B-spline functions form a partition of unity on the interval $[t_n, t_{m+1}]$:

$$\sum_{i=0}^m N_i^n(t) \equiv 1 \quad \forall t \in [t_n, t_{m+1}].$$

Some Properties of B-spline Basis Functions

Piecewise polynomials and smoothness

- ▶ $N_i^k(t)$ is a piecewise polynomial, each polynomial piece being of degree k .
- ▶ At a knot of multiplicity r , $1 \leq r \leq k$, it is $\mathcal{C}^{k-r}$ continuous.
- ▶ In particular, a simple knot gives $\mathcal{C}^{k-1}$ continuity, whereas a knot of multiplicity k gives only $\mathcal{C}^0$ continuity.

Non-negativity and local support

- ▶ $N_i^k(t) \geq 0$ for all t .
- ▶ $N_i^k(t) = 0$ if $t \notin [t_i, t_{i+k+1}]$
- ▶ $N_i^k(t) > 0$ if $t \in]t_i, t_{i+k+1}[$.
- ▶ On every non-degenerate knot interval, at most $k + 1$ basis functions are nonzero.

Non-uniform rational B-spline (NURBS) curves

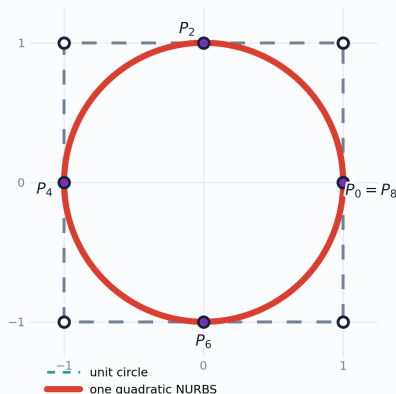
Definition

A degree n NURBS curve with $m + 1$ control points is expressed as

$$\mathbf{P}(t) = \frac{\sum_{i=0}^m w_i \mathbf{P}_i N_i^n(t)}{\sum_{i=0}^m w_i N_i^n(t)}.$$

- ▶ A **weight** is introduced for each control point.
- ▶ It is the central projection of a B-spline curve in one more dimension.
- ▶ Most algorithms for B-spline curves extend to NURBS curves (similarly to the Bézier case).

NURBS: An Exact Circle



A quadratic NURBS

9 control points

Knot vector:

$[0, 0, 0, 1, 1, 2, 2, 3, 3, 4, 4, 4]$

Each knot span is a rational quadratic circular arc.

Local control + exact conics.